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W3

Q1. (a) $\vdash_{PL} P \to (P \to P)$

This is just an instance of PL1

$P \to (P \to P)$ (PL1) ✓

(b) $\vdash_{PL} P \to P$

✓

1. $(P \to ((P \to P) \to P)) \to ((P \to (P \to P)) \to (P \to P))$ (PL2)
2. $P \to ((P \to P) \to P)$ (PL1)
3. $(P \to (P \to P)) \to (P \to P)$ MP, 1, 2
4. $P \to (P \to P)$ (PL1)
5. $P \to P$ MP, 3,4

(c) $\vdash_{PL} (\sim P \to P) \to P$

1. $(\sim P \to \sim P) \to ((\sim P \to P) \to P)$ (PL3)
2. $(\sim P \to ((\sim P \to \sim P) \to \sim P)) \to ((\sim P \to (\sim P \to \sim P)) \to (\sim P \to \sim P))$ (PL2)
3. $\sim P \to ((\sim P \to \sim P) \to \sim P)$ PL1
4. $(\sim P \to (\sim P \to \sim P)) \to (\sim P \to \sim P)$ MP, 2,3

5. $(\sim P \to (\sim P \to \sim P))$ PL1

6. $\sim P \to \sim P$ MP, 4, 5

7. $(\sim P \to P) \to P$ MP, 1, 6 ✓

Q2. (a) $\vdash_{PL} \phi \to ((\phi \to \psi) \to \psi)$

We first prove $\phi, \phi \to \psi \vdash \psi$ ✓

1. ϕ premise
2. $\phi \to \psi$ premise
3. ψ MP, 1, 2

By the deduction theorem

$\phi \vdash_{PL} (\phi \to \psi) \to \psi$

By the deduction theorem again

$\vdash_{PL} \phi \to ((\phi \to \psi) \to \psi)$

(b) $\phi \vdash_{PL} \psi \to \phi$

We first prove $\phi, \psi \vdash_{PL} \phi$ ✓

1. ϕ premise

By DT, $\phi \vdash_{PL} \psi \to \phi$

(c) $\vdash_{PL} \sim\phi \to (\phi \to \psi)$

We first prove

$\sim\phi, \phi \vdash_{PL} \psi$

1. $\phi \to (\sim\phi \to \psi)$ PL1
2. $(\phi \to (\sim\phi \to \psi)) \to ((\phi \to \sim\phi) \to (\phi \to \psi))$ PL2
3. $((\phi \to \sim\phi) \to (\phi \to \psi))$ PL3
4. $\sim\phi \to (\phi \to \sim\phi)$ PL1
5. $\sim\phi$ premise
6. $\phi \to \sim\phi$ MP, 4, 5
7. $\phi \to \psi$ MP, 3, 6
8. ϕ premise
9. ψ

✓

By DT, $\sim\phi \vdash_{PL} \phi \to \psi$

By DT, $\vdash_{PL} \sim\phi \to (\phi \to \psi)$

(d) $\vdash_{PL} \sim\sim\phi \to \phi$

Prove $\sim\sim\phi \vdash_{PL} \phi$ first

1. $\sim\sim\phi \to (\sim\phi \to \sim\sim\phi)$ PL1
2. $(\sim\phi \to \sim\sim\phi) \to ((\sim\phi \to \sim\phi) \to \phi)$ PL2
3. $\sim\sim\phi$ premise
4. $\sim\phi \to \sim\sim\phi$ MP, 1, 3
5. $(\sim\phi \to \sim\phi) \to \phi$ MP, 2, 4
6. $(\sim\phi \to ((\sim\phi \to \sim\phi) \to \sim\phi)) \to ((\sim\phi \to (\sim\phi \to \sim\phi)) \to (\sim\phi \to \sim\phi))$ PL2
7. $\sim\phi \to ((\sim\phi \to \sim\phi) \to \sim\phi)$ PL1
8. $(\sim\phi \to (\sim\phi \to \sim\phi)) \to (\sim\phi \to \sim\phi)$ MP, 6, 7
9. $\sim\phi \to (\sim\phi \to \sim\phi)$ PL1
10. $\sim\phi \to \sim\phi$ MP, 8, 9
11. ϕ MP, 5, 10

By DT, $\vdash_{PL} \sim\sim\phi \to \phi$

✓

(e) $\vdash_{PL} \phi \to \sim\sim\phi$

From (d), we know that $\vdash_{PL} \sim\sim\phi \to \phi$

We also want to prove $\vdash_{PL} (\sim\psi \to \sim\phi) \to (\phi \to \psi)$

First, prove $\sim\psi \to \sim\phi, \phi \vdash_{PL} \psi$

1. $\sim\psi \to \sim\phi$ premise
2. $(\sim\psi \to \sim\phi) \to ((\sim\psi \to \phi) \to \psi)$, PL2
3. $(\sim\psi \to \phi) \to \psi$ MP, 1,2
4. ϕ premise
5. $\phi \to (\sim\psi \to \phi)$ PL1
6. $\sim\psi \to \phi$ MP, 4, 5
7. ψ MP, 3, 6

By DT, $\vdash_{PL} (\sim\psi \to \sim\phi) \to (\phi \to \psi)$

Now, we want to show that $(\sim\psi \to \sim\phi) \to (\phi \to \psi), \sim\sim\phi \to \phi \vdash_{PL} \phi \to \sim\sim\phi$

1. $\sim\sim\sim\phi \to \sim\phi$ premise
2. $(\sim\sim\sim\phi \to \sim\phi) \to (\phi \to \sim\sim\phi)$ premise
3. $\phi \to \sim\sim\phi$ MP, 1,2

By cut, if $\Gamma \vdash_{PL} \delta$ and $\Sigma, \delta \vdash_{PL} \phi$, then $\Sigma, \Gamma \vdash_{PL} \phi$

Since $\vdash_{PL} \sim\sim\phi \to \phi$, $(\sim\psi \to \sim\phi) \to (\phi \to \psi), \sim\sim\phi \to \phi \vdash_{PL} \phi \to \sim\sim\phi$

$\therefore (\sim\psi \to \sim\phi) \to (\phi \to \psi) \vdash_{PL} \phi \to \sim\sim\phi$

Again, by cut,

since $\vdash_{PL} (\sim\psi \to \sim\phi) \to (\phi \to \psi)$, $(\sim\psi \to \sim\phi) \to (\phi \to \psi) \vdash_{PL} \phi \to \sim\sim\phi$ ✓

$\therefore \vdash_{PL} \phi \to \sim\sim\phi$ //

Q3 (a) $\Gamma \vdash_{PL}$ iff for any ϕ, $\Gamma \vdash_{PL} \phi$

Lemma (Converse of DT): if $\Gamma \vdash \phi \to \psi$, then $\Gamma \cup \{\phi\} \vdash \psi$

Proof: if $\Gamma \vdash \phi \to \psi$, then by MP, $\phi, \phi \to \psi \vdash \psi$. By cut, $\Gamma, \phi \vdash \psi$ //

($\Rightarrow$) Assume $\Gamma \vdash_{PL}$ ①

From 2c, $\vdash_{PL} \sim\phi \to (\phi \to \psi)$

without loss, we can rewrite this as

$\vdash_{PL} \sim(\phi \to \phi) \to ((\phi \to \phi) \to \phi)$

By Converse of DT, $\sim(\phi \to \phi) \vdash_{PL} (\phi \to \phi) \to \phi$ ✓

$\therefore \bot \vdash_{PL} (\phi \to \phi) \to \phi$ ②

From 1(b), $\vdash_{PL} P \to P$

we can prove this for any ϕ

$\phi \vdash_{PL} \phi$

1. ϕ premise

By DT, $\vdash_{PL} \phi \to \phi$ ③

By Cut, by using ①, ②, ③, we reach $\Gamma \vdash_{PL} \phi$ for any ϕ

($\Leftarrow$) Assume for any ϕ, $\Gamma \vdash_{PL} \phi$

Then, trivially, we can let ϕ be $\sim(P \to P)$

$\therefore \Gamma \vdash_{PL} \bot$ ✓

$\therefore \Gamma \vdash_{PL}$

(b) $\Gamma, \sim\phi \vdash_{PL}$ iff $\Gamma \vdash_{PL} \phi$

($\Rightarrow$) Assume $\Gamma, \sim\phi \vdash_{PL}$

This does not mean $\Gamma, \neg\varphi \vdash \neg(\varphi \to \varphi)$ for any φ, but $\Gamma, \neg\varphi \vdash \neg(\rho \to \rho)$ for a fixed ρ

By DT,

$$\Gamma \vdash_{PL} \sim\phi \to \sim(\phi \to \phi)$$

Now, we want to show that $(\sim\psi \to \sim\phi) \vdash_{PL} (\phi \to \psi)$

First, prove $\sim\psi \to \sim\phi, \phi \vdash_{PL} \psi$

1. $\sim\psi \to \sim\phi$ premise
2. $(\sim\psi \to \sim\phi) \to ((\sim\psi \to \phi) \to \psi)$, PL2
3. $(\sim\psi \to \phi) \to \psi$ MP, 1,2
4. ϕ premise
5. $\phi \to (\sim\psi \to \phi)$ PL1
6. $\sim\psi \to \phi$ MP, 4, 5
7. ψ MP, 3, 6

By DT, $\sim\psi \to \sim\phi \vdash_{PL} \phi \to \psi$

Then, substituting ϕ for ψ & $(\phi \to \phi)$ for ϕ

$\therefore \sim\phi \to \sim(\phi \to \phi) \vdash_{PL} (\phi \to \phi) \to \phi$

By Cut, $\Gamma \vdash_{PL} (\phi \to \phi) \to \phi$

Since $\vdash_{PL} \phi \to \phi$ from 1b,

by cut again, $\Gamma \vdash_{PL} \phi$ ///

($\Leftarrow$) Assume that $\Gamma \vdash_{PL} \phi$

We want to show that

$\phi, \sim\phi \vdash_{PL}$

1. ϕ premise
2. $\sim\phi$ premise
3. $\phi \to (\sim(\phi \to \phi) \to \phi)$ PL1
4. $\sim(\phi \to \phi) \to \phi$ MP, 1, 3
5. $\sim\phi \to (\sim(\phi \to \phi) \to \sim\phi)$ PL1
6. $\sim(\phi \to \phi) \to \sim\phi$ MP, 2, 5
7. $(\sim(\phi \to \phi) \to \sim\phi) \to ((\sim(\phi \to \phi) \to \phi) \to \sim(\phi \to \phi))$ PL3
8. $(\sim(\phi \to \phi) \to \phi) \to \sim(\phi \to \phi)$ MP, 6, 7
9. $\sim(\phi \to \phi)$ MP, 5, 8

Again, you should show $\neg(P \to P)$, not $\neg(Q \to Q)$

By Cut, $\Gamma, \sim\phi \vdash_{PL}$ $\#$

(c) $\Gamma, \phi \vdash_{PL}$ iff $\Gamma \vdash_{PL} \sim\phi$

($\Leftarrow$) Assume $\Gamma \vdash_{PL} \sim\phi$.

This direction is the same as in (b)

We proved that $\phi, \sim\phi \vdash_{PL}$

By cut, $\Gamma, \phi \vdash_{PL}$

($\Rightarrow$) Assume $\Gamma, \phi \vdash_{PL}$

By DT, $\Gamma \vdash \phi \to \sim(\phi \to \phi)$

Here as well

Again, in (b), we showed that

$$\sim\psi \to \sim\phi \vdash_{PL} \phi \to \psi$$

letting ψ be $\sim\phi$ & ϕ be $(\phi \to \phi)$

we have

$$\sim\sim\phi \to \sim(\phi \to \phi) \vdash_{PL} (\phi \to \phi) \to \sim\phi$$

We want to show that

$$\phi \to \sim(\phi \to \phi) \vdash_{PL} \sim\sim\phi \to \sim(\phi \to \phi)$$

1.	$\sim\sim\phi$	premise
2.	$\phi \to \sim(\phi \to \phi)$	premise
3.	$\sim\sim\phi \to \phi$	Theorem from 2 d
4.	ϕ	MP 1, 3
5.	$\sim(\phi \to \phi)$	MP, 2, 4

By DT, $\phi \to \sim(\phi \to \phi) \vdash_{PL} \sim\sim\phi \to \sim(\phi \to \phi)$

By Cut, $\phi \to \sim(\phi \to \phi) \vdash_{PL} (\phi \to \phi) \to \sim\phi$

By Cut, $\Gamma \vdash_{PL} (\phi \to \phi) \to \sim\phi$

From 1b, $\vdash_{PL} (\phi \to \phi)$

By DT & Cut, $\Gamma \vdash_{PL} \phi$ ///

Q4. (a) $\vdash_T \Box\Box P \to P$

1. $\Box P \to P$ (T)
2. $\Box(\Box P \to P)$ (NEC)
3. $\Box(\Box P \to P) \to (\Box\Box P \to \Box P)$ (K)
4. $\Box\Box P \to \Box P$ MP 2,3
5. $((\Box\Box P \to \Box P) \to (\Box P \to P)) \to (\Box\Box P \to P)$ (PL)
6. $\Box\Box P \to P$ MP 4, PL, MP 1, PL

(b) $\vdash_T P \to \Diamond P$

Prove $\vdash_T P \to \sim\Box\sim P$

1. $\Box\sim P \to \sim P$ T
2. $(\Box\sim P \to \sim P) \to (P \to \sim\Box\sim P)$ PL
3. $P \to \sim\Box\sim P$ MP, 1,2

$\therefore \vdash_T P \to \Diamond P$

(c) $\vdash_{S4} \Diamond\Diamond\Diamond P \to \Diamond P$

which is $\vdash_{S4} \sim\Box\sim\sim\Box\sim\sim\Box\sim P \to \sim\Box\sim P$

1. $\Box\sim\sim\Box\sim P \to \Box\Box\sim\sim\Box\sim P$ S4
2. $\Box\sim\sim\Box\sim P \to \sim\sim\Box\sim\sim\Box\sim P$ PL
3. $\Box(\Box\sim\sim\Box\sim P \to \sim\sim\Box\sim\sim\Box\sim P)$ NEC, 2
4. $\Box(\Box\sim\sim\Box\sim P \to \sim\sim\Box\sim\sim\Box\sim P) \to (\Box\Box\sim\sim\Box\sim P \to \Box\sim\sim\Box\sim\sim\Box\sim P)$ K
5. $\Box\Box\sim\sim\Box\sim P \to \Box\sim\sim\Box\sim\sim\Box\sim P$ MP 3,4
6. $\Box\sim\sim\Box\sim P \to \Box\sim\sim\Box\sim\sim\Box\sim P$ PL (syllogism), 1, 5
7. $\sim\Box\sim\sim\Box\sim\sim\Box\sim P \to \sim\Box\sim\sim\Box\sim P$ PL (contraposition), 6
8. $\Box\sim P \to \Box\Box\sim P$ S4

9. $\Box\sim P \to \sim\sim\Box\Box\sim P$ PL

10. $\Box\Box\sim P \to \Box\sim\sim\Box\sim P$ 9, NEC, K, MP

11. $\Box\sim P \to \Box\sim\sim\Box\sim P$ PL (syllogism) 8, 10

12. $\sim\Box\sim\sim\Box\sim P \to \sim\Box\sim P$ PL (contraposition) 11 ✓

13. $\sim\Box\sim\sim\Box\sim\sim\Box\sim P \to \sim\Box\sim P$ PL (syllogism) 7, 12

14. $\Diamond\Diamond\Diamond P \to \Diamond P$ def

Alternatively,

1. $\Diamond\Diamond\Diamond P \to \Diamond\Diamond P$ (S4$\Diamond$)
2. $\Diamond\Diamond P \to \Diamond P$ (S4$\Diamond$)
3. $\Diamond\Diamond\Diamond P \to \Diamond P$ PL (syllogism) 1, 2

(d) $\vdash_{S4} \Box P \to \Box\Diamond\Box P$

i.e. $\vdash_{S4} \Box P \to \Box\sim\Box\sim\Box P$

1. $\Box P \to \Diamond\Box P$ T$\Diamond$
2. $\Box\Box P \to \Box\Diamond\Box P$ Becker 1
3. $\Box P \to \Box\Box P$ S4
4. $\Box P \to \Box\Diamond\Box P$ PL (syllogism) 3, 2 ✓

Proof of $\Diamond$T

1. $\Box\sim\phi \to \sim\phi$ T
2. $\phi \to \sim\Box\sim\phi$ PL (contraposition), 1

$\therefore \vdash_T \phi \to \Diamond\phi$

(e) $\vdash_{S5} \Box(\Box P \to \Box Q) \vee \Box(\Box Q \to \Box P)$

Lemma 1: The B$\Diamond$-axiom is provable in S5

$\vdash_{S5} \phi \to \Box\Diamond\phi$

1. $\Box\sim\phi \to \sim\phi$ T
2. $\phi \to \Diamond\phi$ PL
3. $\Diamond\phi \to \Box\Diamond\phi$ S5$\Diamond$
4. $\phi \to \Box\Diamond\phi$ PL

Lemma 2: The S4-axiom is provable in S5

$\vdash_{S5} \Box\phi \to \Box\Box\phi$

1. $\Box\phi \to \Box\Diamond\Box\phi$ B$\Diamond$
2. $\Diamond\Box\phi \to \Box\phi$ S5
3. $\Box\Diamond\Box\phi \to \Box\Box\phi$ Becker
4. $\Box\phi \to \Box\Box\phi$ PL

Main proof

1. $\sim\Box P \to (\Box P \to \Box Q)$ PL (2.)
2. $\Box\sim\Box P \to \Box(\Box P \to \Box Q)$ Becker, 1
3. $\sim\Box(\Box P \to \Box Q) \to \Diamond\Box P$ PL, 2
4. $\Diamond\Box P \to \Box P$ S5

5. $\sim\Box(\Box P \to \Box Q) \to \Box P$ PL, 3, 4
6. $\Box P \to \Box\Box P$ S4

7. $\sim\Box(\Box P \to \Box Q) \to \Box\Box P$ PL, 5, 6 ✓
8. $\Box P \to (\Box Q \to \Box P)$ PL1
9. $\Box\Box P \to \Box(\Box Q \to \Box P)$ Becker, 8
10. $\sim\Box(\Box P \to \Box Q) \to \Box(\Box Q \to \Box P)$ PL, 7, 9
11. $\Box(\Box P \to \Box Q) \vee \Box(\Box Q \to \Box P)$ def of $\vee$

Q5. (a) i. $\vdash_{K\Diamond} \Diamond(\phi \wedge \psi) \to \Diamond\phi \wedge \Diamond\psi$

1. $\phi \wedge \psi \to \phi$ PL
2. $\Box(\phi \wedge \psi \to \phi)$ NEC 1
3. $\Box(\phi \wedge \psi \to \phi) \to (\Diamond(\phi \wedge \psi) \to \Diamond\phi)$ $K\Diamond$
4. $\Diamond(\phi \wedge \psi) \to \Diamond\phi$ MP
5. $\phi \wedge \psi \to \psi$ PL
6. $\Box(\phi \wedge \psi \to \psi)$ NEC 5
7. $\Box(\phi \wedge \psi \to \psi) \to (\Diamond(\phi \wedge \psi) \to \Diamond\psi)$ $K\Diamond$
8. $\Diamond(\phi \wedge \psi) \to \Diamond\psi$ MP
9. $\Diamond(\phi \wedge \psi) \to \Diamond\phi \wedge \Diamond\psi$ PL and a clever use of DT

ii. $\vdash_{K\Diamond} \Box(\phi \to \psi) \to (\Box\phi \to \Box\psi)$

1. $(\phi \to \psi) \to (\sim\psi \to \sim\phi)$ PL
2. $\Box(\phi \to \psi) \to \Box(\sim\psi \to \sim\phi)$ 1, NEC, K, MP
3. $\Box(\sim\psi \to \sim\phi) \to (\Diamond\sim\psi \to \Diamond\sim\phi)$ K ✓
4. $\Box(\phi \to \psi) \to (\Diamond\sim\psi \to \Diamond\sim\phi)$ PL
5. $\Box(\phi \to \psi) \to (\sim\Diamond\sim\phi \to \sim\Diamond\sim\psi)$ PL (contraposition)

By substitution of equivalent formulas ($\Diamond$ df)

since $\sim\Diamond\sim\phi \leftrightarrow \Box\phi$

$\vdash_{K\Diamond} \Box(\phi \to \psi) \to (\Box\phi \to \Box\psi)$

$\therefore \vdash_{K\Diamond} \Box(\phi \to \psi) \to (\Diamond \phi \to \Diamond \psi)$ $\#$ ← I've normally seen this symbol for contradiction. When finishing a proof, either □ or q.e.d is more normal

(b) Claim: for any MPL-wff ϕ, $\vdash_K \phi$ iff $\vdash_{K\Diamond} \phi$

($\Rightarrow$) Assume $\vdash_K \phi$. Then, there is proof s.t.

$$\begin{array}{c} \psi_1 \\ \psi_2 \\ \vdots \\ \psi_n \\ \hline \phi \end{array}$$

where ψ_i is a K-axiom, or ψ_i follows from earlier wffs via a K-rule. Suppose we try to prove ϕ in the K$\Diamond$ variant axiomatisation. Then, from (ii), we know that $\vdash_{K\Diamond} \Box(\phi \to \psi) \to (\Box\phi \to \Box\psi)$, $\vdash_{K\Diamond} \sim\Box\sim\phi \leftrightarrow \Diamond\phi$
$\therefore$ for any ψ_i that is an MPL-instance of a K-axiom, it can be derived from an MPL instance of a K$\Diamond$ axiom. Since K$\Diamond$ retains PL1-3 axioms, so, $\vdash_{K\Diamond} \phi$

($\Leftarrow$) Assume $\vdash_{K\Diamond} \phi$. Then, there is a proof s.t.

$$\begin{array}{c} \psi_1 \\ \psi_2 \\ \vdots \\ \psi_n \\ \hline \phi \end{array}$$

✓

We know that $\vdash_K \sim\Diamond\sim\phi \leftrightarrow \Box\phi$ and $\vdash_K \Box(\phi \to \psi) \to (\Box\phi \to \Box\psi)$. Suppose we try to prove ϕ in the K-axiomatisation. Then, for any ψ_i that is an instance of the K$\Diamond$ axiom, we can derive it from an instance of a K axiom. Since K also retains PL1-3 axioms from K$\Diamond$, so, $\vdash_k \phi$